

Free-slip immersed boundaries for a collocated SIMPLEC solver

Two routes: ghost extrapolation and partial face areas

Notes generated during solver development

September 4, 2026

Abstract

We extend the implicit second-order immersed-boundary method of Luchini, Gatti, Chiarini, Gattere, Atzori & Quadrio (*J. Comput. Phys.* **539** (2025) 114245) from no-slip to *free-slip* boundaries, inside a collocated finite-volume SIMPLEC solver. The original method folds the entire boundary correction into the centre weight of the Laplacian stencil. We show that free slip is obtained by replacing the Dirichlet wall value with the *tangential projection* of the first fluid point, which keeps the correction in the same place but introduces a coupling between the velocity components. We then document three distinct $O(1)$ errors that appear if the construction is applied only partially, and two instabilities that appear if it is applied naively, together with the measurements that identify each. The final scheme reproduces uniform flow along an oblique free-slip wall to 2×10^{-6} where the staircase alternative is wrong by 1.2×10^{-1} , and runs at every resolution tested on both a coupled and a segregated momentum route.

We then present a second, independent route based on *partial face areas* (cut cells). There the boundary condition is carried entirely by a geometric closure identity, so no ghost value, no blanking and no cross term are needed at all. It reproduces the oblique-wall test exactly ($g_x = 0$ to the last bit) and makes the discrete continuity error machine-zero in both the fluid and the solid, which the ghost route does not achieve. We document both, and say plainly which to prefer and why.

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Part I — The ghost-extrapolation route

1 Governing equations and baseline discretisation

We solve the incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla p + \mathbf{g}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

with p the reduced pressure, on a uniform doubly periodic Cartesian grid. The discretisation is cell-centred (collocated): u , v and p all live at cell centres, and the five-point stencil is stored as five coefficient arrays rather than as a matrix. Time advance is implicit Euler used as pseudo-transient continuation, and the pressure–velocity coupling is SIMPLEC exactly as in OpenFOAM’s `simpleFoam`:

$$a_P \mathbf{u}_P = H(\mathbf{u}) - V \nabla p, \quad H(\mathbf{u}) = b - \sum_{nb} a_{nb} \mathbf{u}_{nb}, \quad (2)$$

$$r_{AU} = 1/a_P, \quad r_{AtU} = 1/\left(a_P + \sum_{nb} a_{nb}\right), \quad (3)$$

$$\nabla \cdot (r_{AtU} V \nabla p) = \nabla \cdot \phi_{HbyA}, \quad \phi = \phi_{HbyA} - r_{AtU} V \text{snGrad}(p) |\mathbf{S}_f|. \quad (4)$$

1.1 Constant flow rate

Free slip generates no friction, so a fixed body force gives no steady state: the flow accelerates without bound. We therefore fix the bulk velocity and let the force follow. Because only the source term depends on g_x , and one predictor–projection pass is *affine* in it, two passes — one at $g_x = 0$ and one at $g_x = 1$ — give the exact g_x that lands the bulk velocity on its target:

$$g_x = \frac{U_b^{\text{target}} - U_b^{(0)}}{U_b^{(1)} - U_b^{(0)}}. \quad (5)$$

Ghost value: $u_{\text{ghost}} = a_U u_P + b v_P$, $a_U = 1 - (h/\delta)n_x^2$, $b = -(h/\delta)n_x n_y$

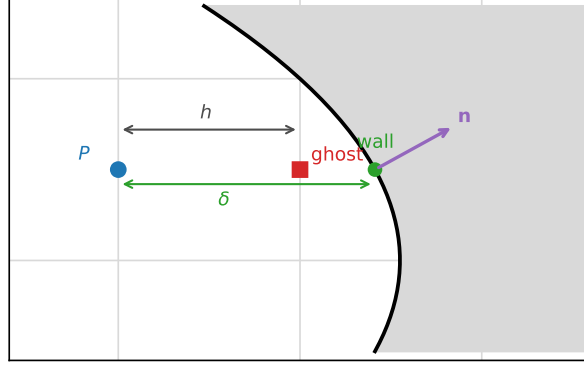


Figure 1: Ghost construction on a cut stencil arm. P is the first fluid point, δ the distance to the true boundary, h the mesh spacing, $\mathbf{n}$ the local surface normal. For no slip the wall value is zero; for free slip it is the tangential projection of $\mathbf{u}_P$.

There is no gain to tune and both passes share the matrix factorisations. A proportional controller was tried first and diverged, because free slip leaves the bulk mode almost undamped.

2 The no-slip starting point

For no slip, Eq. (6) of Luchini *et al.* replaces the ghost value at the first point inside the solid by a linear extrapolation between the first fluid point and the wall,

$$u_{\text{ghost}} = \left(1 - \frac{h}{\delta}\right) u_P, \quad (6)$$

with h the mesh spacing and δ the distance from the cell centre to the wall along that stencil arm (Fig. 1). Substituting back into the Laplacian moves the whole correction into the *centre* weight, so nothing is stored at the ghost point and corrections from different directions are additive. Crucially the correction *increases* diagonal dominance as $\delta \rightarrow 0$, which is what makes the method robust.

3 Free slip: the tangential projection

Free slip means no penetration *and* zero tangential gradient. The second condition says the tangential velocity does not change along the normal, so to leading order the wall velocity equals the tangential part of the first fluid point:

$$\mathbf{u}_{\text{wall}} = (I - \mathbf{n}\mathbf{n}^T) \mathbf{u}_P, \quad (7)$$

which is precisely the single scalar constraint $\mathbf{u}_{\text{wall}} \cdot \mathbf{n} = 0$ written out. Feeding (7) through the same linear extrapolation that produced (6) gives, for the x -momentum,

$$u_{\text{ghost}} = a_U u_P + b v_P, \quad a_U = 1 - \frac{h}{\delta} n_x^2, \quad b = -\frac{h}{\delta} n_x n_y \quad (8)$$

and symmetrically $a_V = 1 - (h/\delta)n_y^2$ for the y -momentum. Setting the projector to the identity ($n_x^2 \rightarrow 1$, $n_x n_y \rightarrow 0$) recovers (6) exactly; we verified that the resulting stencil coefficients are bitwise identical to the no-slip ones.

So the centre weight picks up n_x^2 in place of 1, and a **cross term** appears in the other component. Everything else about the method is preserved: one value per cut arm, no stored ghosts, additive across directions.

3.1 Why a single constraint per cell matters

An earlier version imposed no penetration once per cut *arm*: $u = 0$ on x -arms and $v = 0$ on y -arms. On a curved boundary a cell can have both an x - and a y -arm cut, and both components are then pinned — the cell becomes a no-slip patch. Measured over five grids, the surface slip velocity decayed as $|u_t| \propto \Delta$: the scheme converged to *no slip*. Equation (7) removes this, because a cell whose normal is mostly along y gets $n_x^2 \approx 0$ on its x -arm too, leaving u free.

4 Three places the ghost must be used

The ghost is not a Laplacian-only correction. Free slip fails, each time at $O(1)$, if it is omitted from any of three places.

(i) **The viscous stencil.** As derived above.

(ii) **The face fluxes.** Closing cut faces on the staircase (a boolean mask) blocks the tangential flux, because for an oblique wall the tangential velocity has a component normal to every axis face. Using the ghost instead, a uniform tangential field gives $u_{\text{ghost}} = u_P$ — the projector is the identity on such a field — so every one of a cut cell’s four *full* faces carries the uniform flux and the discrete divergence is identically zero. Measured on a 45° wall: $\max |\nabla \cdot \phi|$ over fluid cells is 1.2×10^{-1} with masks and 4.4×10^{-10} with the ghost.

(iii) **The convective stencil.** The original paper notes that no correction of the convective terms is needed for second order. That is true *for no slip*, where $u_P \rightarrow 0$ at the wall makes the term negligible. Under free slip u_P is $O(U)$, and reading the blanked zero out of the solid neighbour is an $O(1)$ error. Section 7 quantifies it.

5 Two instabilities, and their cure

5.1 The cross term must not be indefinite

Collecting the ghost contributions, each cut arm adds to the local 2×2 block

$$\frac{h}{\delta} \mathbf{n} \mathbf{n}^T \times \begin{cases} +\nu & (\text{viscous}) \\ \frac{1}{2}f & (\text{convective, central}) \end{cases} \quad (9)$$

where f is the face volume flux. Since $\mathbf{n} \mathbf{n}^T$ is rank-one positive semi-definite, the viscous block is $a_P I + \nu \frac{h}{\delta} \mathbf{n} \mathbf{n}^T$, with eigenvalues $a_P + \nu h/\delta$ along $\mathbf{n}$ and a_P along $\mathbf{t}$: well posed, constrained in the normal direction only — which is exactly what free slip means — and increasingly dominant as $\delta \rightarrow 0$ (Fig. 2, left). The convective copy carries f , which *changes sign*, so that block is indefinite (Fig. 2, right). Making an indefinite block implicit does not make it well posed: a central convective ghost diverged whether it was lagged, lagged with a diagonal repair, clipped in h/δ , or solved fully implicitly.

Cure: upwind the cut arms. The ghost is a *neighbour* value, so it should only be picked up on inflow faces. With upwinding the inflow contribution becomes

$$f \left(I - \frac{h}{\delta} \mathbf{n} \mathbf{n}^T \right) = -|f|I + |f| \frac{h}{\delta} \mathbf{n} \mathbf{n}^T, \quad f < 0, \quad (10)$$

whose rank-one part is positive semi-definite again, while the $-|f|I$ part is balanced by $+f$ on the outflow faces exactly as in ordinary upwinding. Uniform flow parallel to the wall remains exact, because upwind and central agree for a constant field. Only cut arms are upwinded; the scheme stays central everywhere else.

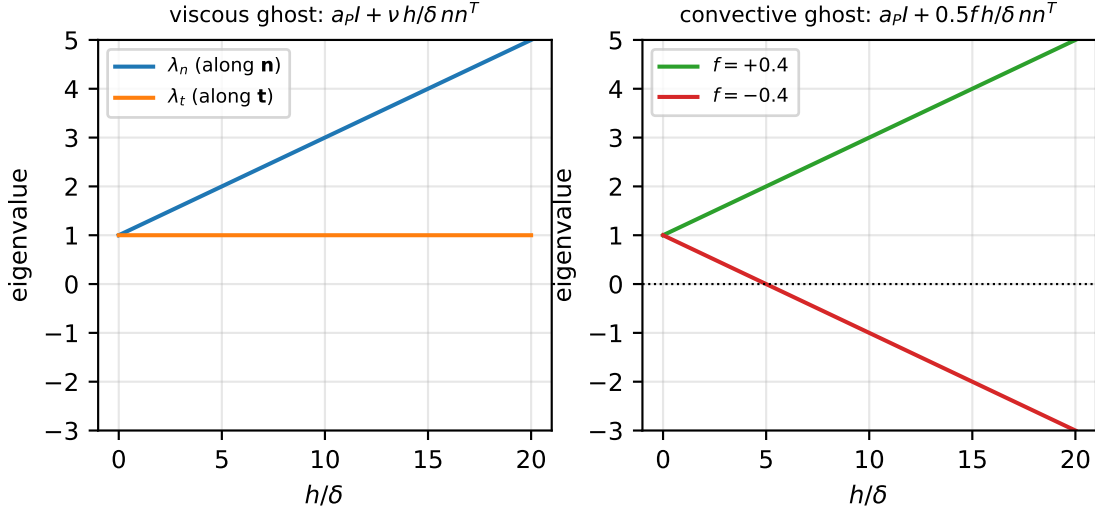


Figure 2: Eigenvalues of the local 2×2 momentum block against h/δ . Left: the viscous ghost is positive semi-definite and stabilising. Right: the convective ghost carries the face flux f , and for $f < 0$ the block loses positivity.

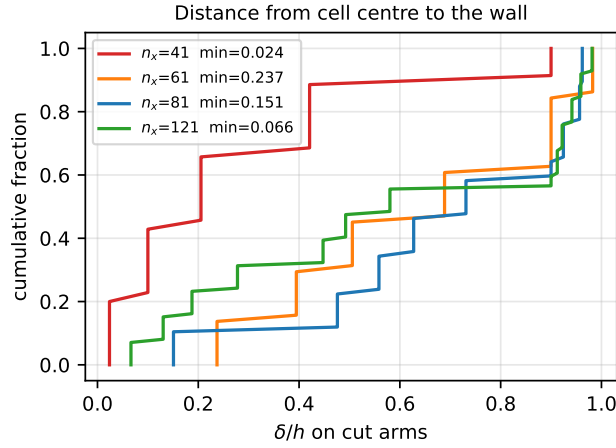


Figure 3: Cumulative distribution of δ/h over cut arms for the circular cylinder. The minimum is a geometric accident of how the circle meets the grid, which is why $n_x = 41$ fails first rather than the coarsest or finest mesh.

5.2 The remaining positivity defect, and the S1/S3 stabilisation

Upwinding is not quite sufficient. On a cut arm the inflow term has *no off-diagonal to land on* — the neighbour is solid — so it falls on the diagonal, and on a few cells very close to the wall it pushes the diagonal non-positive. At $n_x = 41$ this happened on two cells, which drove $r_{A_t U}$ to -121 , turned the pressure Laplacian indefinite, and produced an exactly singular factorisation. The distribution of δ/h (Fig. 3) explains why $n_x = 41$ is the outlier: its closest cut cell sits at $\delta/h = 0.024$ against 0.151 at $n_x = 81$ — a geometric lottery, not a resolution trend.

S1 — floor the SIMPLEC mobilities. r_{A_U} and $r_{A_t U}$ are floored at $V/\Delta t$, the value an unobstructed cell would have, so the pressure Laplacian cannot turn indefinite. This is necessary but not sufficient: with S1 alone the failure changes from a singular factorisation to divergence.

S2 — a global Péclet blend (rejected). Weighting the ghost by $\min(1, C\nu/|f|)$ does restore convergence at every resolution, but it gives up consistency *everywhere* the blend is active in

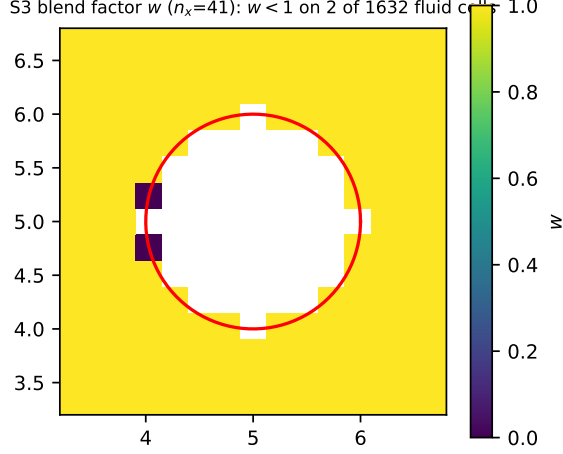


Figure 4: The S3 blend factor w at $n_x = 41$. $w = 1$ (full ghost, fully consistent) everywhere except two cells adjacent to the wall.

order to rescue a handful of cells: the coarse-grid drag moved by 20% as the constant C was varied over a decade. We report it because it works, and discard it because S3 is strictly better.

S3 — an adaptive per-cell blend (adopted). On each cut cell we take

$$\text{contribution} = w (\text{ghost, upwinded}) + (1 - w) (\text{plain central}), \quad (11)$$

with $w \in [0, 1]$ the largest value that keeps the momentum diagonal at or above a floor. The diagonal is linear in w , so w follows in closed form and no iteration is needed. Cells that do not need help keep $w = 1$ and remain fully consistent. At $n_x = 41$ the blend engages on **2 of 1216 fluid cells** (Fig. 4) — precisely the two diagnosed above.

6 Coupled versus segregated momentum, and a Rhie–Chow trap

The cross term couples u and v at the *same* cell only, so the coupled block matrix

$$\begin{bmatrix} A_{uu} & \text{diag}(c_U) \\ \text{diag}(c_V) & A_{vv} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} b_u \\ b_v \end{bmatrix} \quad (12)$$

has six nonzeros per row instead of five — far cheaper than a general block-coupled solve. $H(\mathbf{u})$ then carries the cross contribution, exactly as OpenFOAM’s $H()$ does.

The segregated alternative lags the cross term and adds $|c|$ to the diagonal to keep the lagged iteration diagonally dominant, subtracting it again on the right with the lagged value of the same component so that the fixed point is unchanged. **This repair is not free.** It inflates a_P , which changes r_{AU} and r_{AtU} , and hence the Rhie–Chow flux — a different but self-consistent pressure–velocity coupling. Removing the repair makes the two routes agree exactly (Table 1). Note also that a_P *must* be the repaired diagonal: $H(\mathbf{u})/a_P$ telescopes to $\mathbf{u} + r_{AU} \nabla p$ only if a_P is the diagonal the predictor actually solved with. Using the unrepaired one breaks the identity and produced a negative drag in our tests.

7 Numerical evidence

7.1 Uniform flow along an oblique free-slip wall

The sharpest test available: a free-slip channel at 45° to the grid, driven at fixed flow rate. The exact solution is uniform flow parallel to the walls with *zero* driving force. Table 2 gives the result with and without the convective ghost.

Table 1: Effect of the segregated diagonal repair, $\nu = 0.4$, $n_x = 61$.

	drag	field difference vs coupled
coupled	1.251539	—
segregated, repair removed	1.251539	4.0×10^{-6}
segregated, with repair	1.256236	4.6×10^{-3}

Table 2: 45° free-slip channel, $\nu = 0.4$. Exact answer: $\max u - \min u = 0$ and $g_x = 0$. All cases converged to a residual of 10^{-9} .

route	n_x	ghost on		ghost off	
		$\max u - \min u$	g_x	$\max u - \min u$	g_x
coupled	41	1.9×10^{-6}	-4.9×10^{-9}	1.232×10^{-1}	2.18×10^{-3}
coupled	81	3.4×10^{-6}	-1.0×10^{-8}	1.024×10^{-1}	2.04×10^{-3}
segregated	41	2.2×10^{-6}	-2.2×10^{-8}	1.211×10^{-1}	2.07×10^{-3}
segregated	81	3.4×10^{-6}	-6.3×10^{-8}	9.959×10^{-2}	1.98×10^{-3}
coupled	161	2.9×10^{-6}	-6.8×10^{-9}	8.307×10^{-2}	1.77×10^{-3}
segregated	161	3.7×10^{-6}	-8.3×10^{-8}	8.127×10^{-2}	1.70×10^{-3}

The ghost-off error is not merely large, it is **h-independent**: halving Δ moves it by a factor 1.2. Over two successive refinements ($41 \rightarrow 81 \rightarrow 161$) the observed order of the velocity spread is 0.27 then 0.30, and that of the spurious drag 0.09 then 0.21 (Fig. 5) — flat, not converging. A grid-refinement study therefore shows a tidy convergent sequence — onto the wrong answer. This is the central argument for carrying the ghost into the convective term.

7.2 Cylinder drag

Table 3 gives the drag on a free-slip circular cylinder in a periodic box at fixed flow rate, with the full scheme (upwind ghost, S1, S3), on both momentum routes. All resolutions run stably on both routes.

Table 3: Free-slip cylinder, $Re_R = UR/\nu = 1.25$, $\nu = 0.4$, all converged to 10^{-9} . Drag is $g_x A_{\text{fluid}}$, which balances the body force at steady state.

n_x	41	61	81	121
Δ	0.2439	0.1639	0.1235	0.0826
coupled	1.236070	1.249283	1.247104	1.250610
segregated	1.263816	1.254353	1.250983	1.254389

7.3 Stability coverage

Table 4 records which combinations run. The central convective ghost fails widely; upwinding repairs the segregated route outright and the coupled route down to $n_x = 61$; adding S1 and S3 closes the remaining case.

7.4 The flow field

Figure 7 shows the converged free-slip cylinder with and without the convective ghost, and Fig. 8 the tangential slip velocity around the surface. Without the ghost the near-wall velocity is visibly

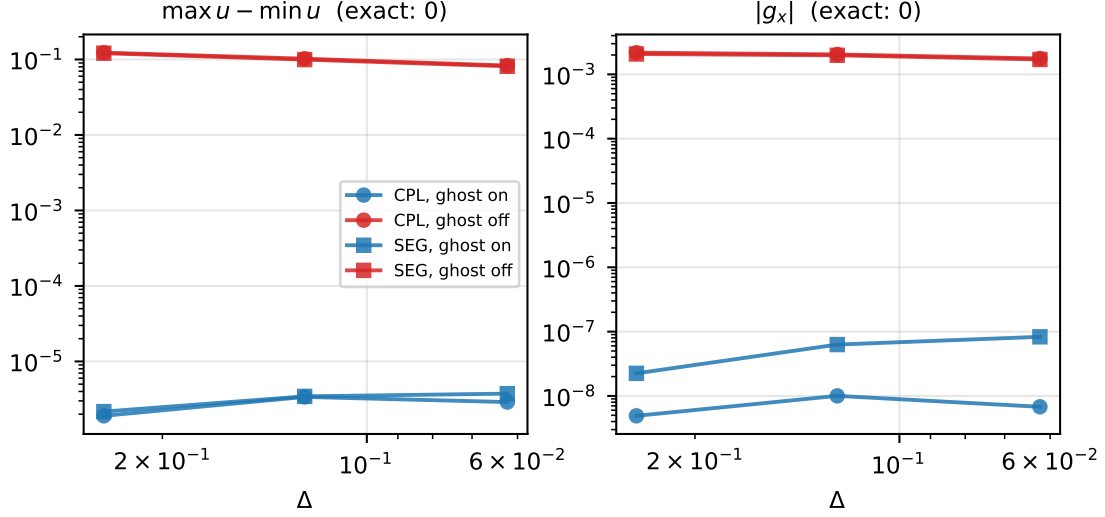


Figure 5: Error on the 45° free-slip channel against mesh spacing. With the ghost the error sits at the solver tolerance; without it the error is essentially independent of Δ .

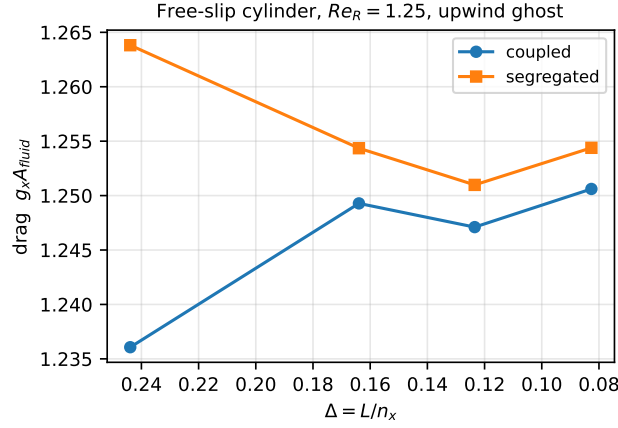


Figure 6: Cylinder drag against mesh spacing for both momentum routes.

suppressed — the boundary behaves partly as no slip, which is the $O(1)$ error of Table 2 seen in physical space.

Part II — The partial-face-area (cut-cell) route

8 Motivation

The ghost route of Part I leaves one defect: continuity is not machine-zero. Removing the face masks makes the body transparent to tangential flow while its interior is still blanked to $\mathbf{u} = 0$, so the ghost extension is discontinuous where it meets the blanked interior, and the resulting divergence source in the first solid layer cannot be removed by the projection because r_{AtU} there is the blanking penalty. It also needs upwinding plus the S1/S3 stabilisation to survive at all resolutions.

Both problems have the same origin: a face is either fully open or fully closed. Giving each

Table 4: Does the case reach a 10^{-9} residual? Cylinder, $\nu = 0.1$, ghost on.

scheme	$n_x = 41$	$n_x = 61$	$n_x = 81$
central ghost, segregated	no	no	yes
central ghost, coupled	no	no	no
upwind ghost, segregated	yes	yes	yes
upwind ghost, coupled	no	yes	yes
upwind ghost + S1 + S3, both	yes	yes	yes

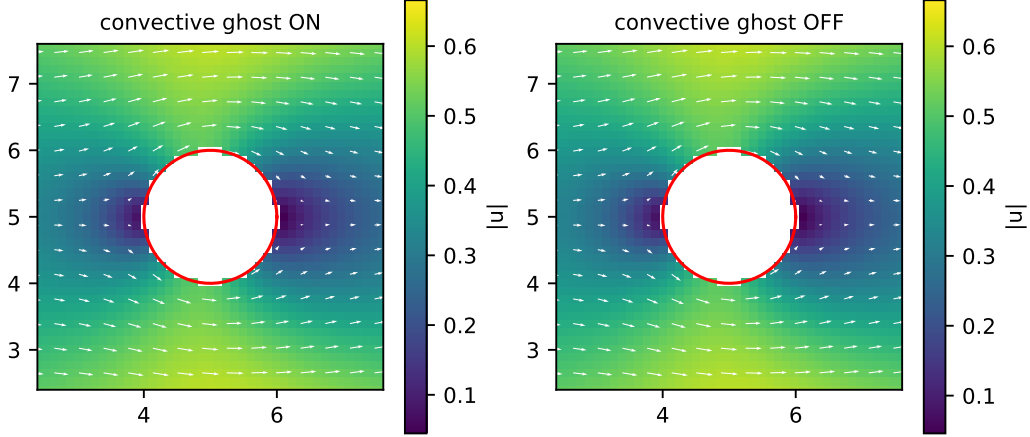


Figure 7: Velocity magnitude and vectors for the free-slip cylinder, $n_x = 81$, $Re_R = 1.25$.

face a *fluid area fraction* removes it.

9 Geometry

Each face carries $\theta \in [0, 1]$, the fraction of the face lying in the fluid, and each cell carries α , the fluid fraction of its volume. Both are obtained from the same level set: θ by bisection along the face, α by sub-sampling. The fluid part of a cut cell is a polygon bounded by the partially open axis faces and the wall segment (Fig. 9), and the two are tied together by

$$\sum_f \theta_f \mathbf{n}_f A_f + \mathbf{n}_{\text{wall}} A_{\text{wall}} = 0. \quad (13)$$

Because θ and the wall segment come from the same level set, (13) holds to machine precision. We verified it directly: on a straight oblique wall the normal recovered from (13) matches the analytic normal to 2.2×10^{-16} , and a uniform field parallel to that wall has *exactly* zero discrete divergence.

10 The boundary condition falls out of the closure

Equation (13) carries the whole free-slip condition, term by term:

- **Viscous.** Free slip means zero tangential stress on the wall segment, and at first order no normal viscous stress either, so the wall contributes nothing: the operator is the θ -weighted face sum, which annihilates a constant field.
- **Convective.** No penetration means no flux through the wall segment, so again only the θ -weighted faces contribute.

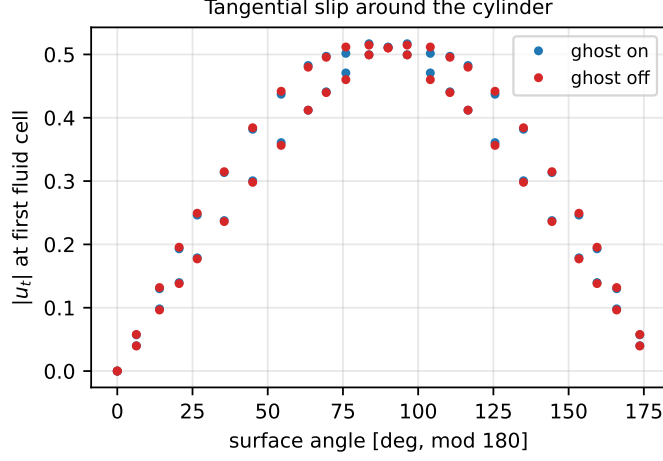


Figure 8: Tangential velocity at the first fluid cell around the cylinder. The convective ghost restores the slip that the blanked-neighbour treatment suppresses.

Cut cell: partially open faces plus the wall segment

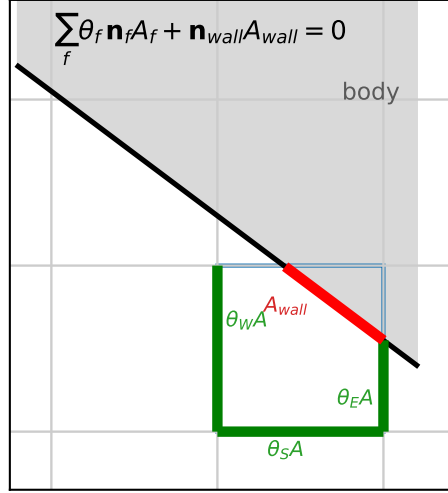


Figure 9: A cut cell. Its fluid boundary consists of the partially open axis faces (green) and the wall segment (red); the two are linked by the closure identity (13).

- **Pressure.** Here the wall *does* contribute — this is the form drag. With $p_{\text{wall}} = p_P$ to first order, (13) turns the Gauss gradient into $\sum_f \theta_f (p_f - p_P) \mathbf{n}_f A_f$, which vanishes for a uniform pressure, so no spurious force is generated on a cut cell.
- **Continuity.** $\sum_f \theta_f \phi_f = 0$. Combined with (13), a uniform velocity in a cut cell satisfies this only if $\mathbf{u} \cdot \mathbf{n}_{\text{wall}} = 0$.

The last point deserves emphasis: **no penetration is never imposed anywhere. It emerges from continuity plus closure.** Consequently this route has no ghost value, no blanking of cut cells, no cross term between the velocity components, and needs none of the S1/S3 stabilisation of Part I. Because the cross term is what distinguished the coupled from the segregated momentum route, the two now produce *bit-identical* results — with no cross coupling they solve the same block-diagonal system. We keep both files only so that the two parts of this note have a matching pair each.

We verified each operator directly against the exact solution of the 45° channel: the viscous residual, the convective residual, the gradient of a uniform pressure and the divergence of the exact field are all 0.000×10^0 .

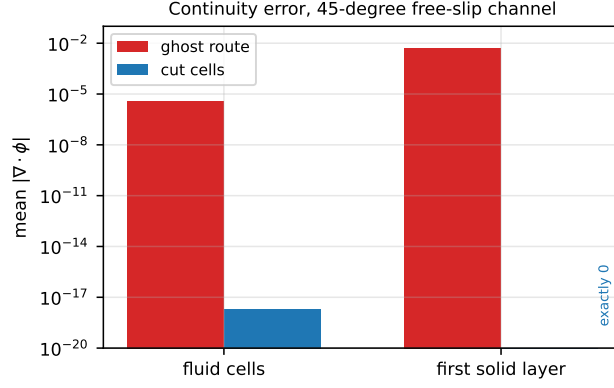


Figure 10: Discrete continuity error on the 45° channel. The cut-cell route is exact in both the fluid and the solid; the ghost route leaves a source in the first solid layer.

11 Results

Table 5: 45° free-slip channel, $\nu = 1$. Exact answer: uniform flow, $g_x = 0$.

	ghost route	cut cells
drag (exact 0)	-4.9×10^{-9}	0.000000
mean $ \nabla \cdot \phi $, fluid	4×10^{-6}	2×10^{-18}
mean $ \nabla \cdot \phi $, first solid layer	4.8×10^{-3}	0

Table 6: Free-slip cylinder, cut-cell route. All cases converged to 10^{-9} , symmetry held to 10^{-14} , continuity machine-zero. Coupled and segregated agree to the last digit.

n_x	41	61	81	121
$\nu = 1.0$ ($Re_R = 0.5$)	2.829667	2.907463	2.938772	2.977790
$\nu = 0.5$ ($Re_R = 1.0$)	1.449213	1.487457	1.503127	1.523026

12 The one cost: small cells

Cut cells with $\alpha \rightarrow 0$ have vanishing volume against $O(h)$ faces — the classic small-cell problem. We absorb slivers with $\alpha < \alpha_{\min}$ into the body. This is the only place the route gives ground, and it has a trap: **an absorbed cell must have its faces closed as well**, or the closure identity (13) is broken for its neighbours and the exactness above is lost. We hit exactly this during development — the divergence of the exact field jumped from 0 to 6×10^{-2} , i.e. $O(h)$, until the faces of absorbed cells were zeroed too. Absorbing moves the boundary by $O(\alpha_{\min} h)$, which is within the first-order error; cell merging would remove even that.

A second trap, worth recording because it is invisible until it is not: on a doubly periodic grid the faces at $x = 0$ and $x = L_x$ are the *same* face. If θ is evaluated independently at both, an inconsistent seam destroys conservation. We compute once and copy. The same applies to the level set used for a test geometry: an oblique wall must be defined through the periodic difference, or it does not close on the torus.

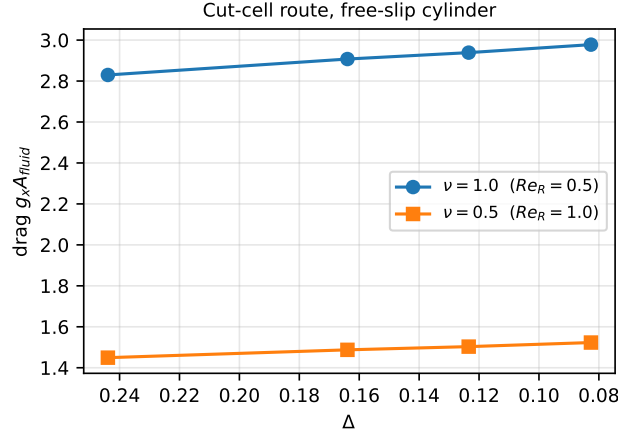


Figure 11: Cut-cell drag against mesh spacing at the two viscosities.

Table 7: The two routes compared.

	ghost (Part I)	cut cells (Part II)
boundary condition	imposed, via the ghost value	emerges from closure
continuity	4×10^{-6} fluid, 5×10^{-3} solid	machine zero
oblique wall	$\sim 10^{-6}$	exact
extra machinery	upwinding, S1, S3, cross term	none
coupled vs segregated	differ (Rhie–Chow, Sec. 7)	bit-identical
weak point	positivity near the wall	small cells
no-slip limit	recovers Eq. (6) exactly	would need the wall stress term

13 Which route to use

For free slip the cut-cell route is clearly preferable: it is exact where the ghost route is approximate, and it needs none of the stabilisation. The ghost route retains one advantage worth keeping in mind — it degenerates *exactly* to Eq. (6) of Luchini *et al.*, so it extends to no slip and to Robin conditions by changing one coefficient, whereas the cut-cell route as written assumes the wall carries no viscous stress and would need that term restored for no slip.

Part III — Limitations common to both

14 Known limitations

Continuity is not machine-zero — in the ghost route only; Part II fixes it. Removing the face masks makes the body transparent to tangential flow while its interior is still blanked to $\mathbf{u} = 0$. The ghost extension is therefore discontinuous where it meets the blanked interior, leaving a divergence source in the first solid layer that the projection cannot remove, because r_{AtU} there is the blanking penalty. Measured: mean $|\nabla \cdot \phi|$ is 4×10^{-6} over fluid cells against 4.8×10^{-3} in the first solid layer. Giving those cells a fluid-like pressure mobility was tried and diverges, because the cut-face coefficient then jumps by orders of magnitude against the small near-wall fluid mobility. The principled cure is **partial face areas** $\theta \in [0, 1]$ in place of the boolean mask — which is exactly Part II, and it does resolve this completely.

Symmetry breaking. The symmetric free-slip state is linearly unstable above $Re_R \approx 5$: the up-down asymmetry grows exponentially from round-off at a constant rate while the solution is otherwise converged, saturating near 50%. No-slip stays symmetric to 10^{-14} at the same Reynolds number. Convergence studies must therefore be run at $\nu \gtrsim 0.4$, where the residual asymmetry is $\sim 10^{-3}$.

First order. Imposing the condition on the staircase-normal arm rather than along the true normal, and evaluating the tangential gradient at the arm rather than along $\mathbf{n}$, are both $O(\Delta)$. The scheme is first-order by construction; the no-slip limit retains the second-order accuracy of the original method.

15 Summary

Free slip in this framework is Eq. (6) of Luchini *et al.* with the Dirichlet wall value replaced by the tangential projection $(I - \mathbf{n}\mathbf{n}^T)\mathbf{u}_P$. The correction stays in the centre weight, but it acquires a cross term between the velocity components, and it must be carried into the face fluxes and the convective stencil as well as the Laplacian — omitting either produces an $O(1)$, mesh-independent error. The convective copy must be upwinded, because it carries the sign-changing face flux and is otherwise indefinite. A small adaptive blend (S3) handles the few cells where the upwind inflow term, having no off-diagonal to occupy, would still destroy positivity.